YUHANG JIANG

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Summary

- EIT Manufacturing double-degree master's in Computer Science (UNITN) and Data Science (SUPSI).
- Two first-author journal papers accepted in 2026: IEEE Transactions on Geoscience and Remote Sensing and TMLR.
- GeoSelect sets the training-free state of the art on both RRSIS-D and RISBench, reaching about 93% of a supervised specialist on RRSIS-D.
- Research intern at Huawei Pisa Research Center and FBK. Kaggle Competitions Expert.
- Work spans embodied AI, mechanistic interpretability of state-space models, adversarial robustness, and remote sensing.

Education

EIT Manufacturing Master & Doctoral School (Double Degree) Sep 2023 - Mar 2026
Data Science and AI for a Competitive Manufacturing
- **Università di Trento (UNITN), Italy**
  MSc, Computer Science · GPA 98/110
- **University of Applied Sciences and Arts of Southern Switzerland (SUPSI), Switzerland**
  Master of Engineering (MSE), Data Science · GPA 5.1/6.0
- **Scholarship:** Recipient of the EIT Manufacturing Scholarship (full tuition waiver + 10,000 EUR/yr)

Anhui University (Project 211), China Sep 2019 - Jun 2023
BEng, Intelligent Science and Technology
- **Average Score:** 84.34/100
- **Awards:** Second Prize for Outstanding Academic Performance (2022); First Prize Scholarship of Academic Science and Technology (2021)
- **Courses:** Machine Learning, Pattern Recognition, Big Data Analysis, Numerical Analysis, Natural Language Processing.

Work Experience

Huawei Pisa Research Center Apr 2026 - Oct 2026
Research Intern Pisa, Italy
Independent research: memory for multimodal agents (first-author paper under review)
- **Controlled experiments located the bottleneck at the delivery step.** Holding the retrieved set fixed and changing one variable at a time, delivering the original images is worth +13.87 points and perfect retrieval only +2.31 on Qwen3-VL-8B; the delivery gain grows on a stronger backbone (+17.92 on Qwen3-VL-32B).
- **Proposed DeliverMem, a training-free delivery method that leaves the memory store unchanged** (original modality, readable identifiers and temporal position, plus a retrieval-side adaptation): **state of the art on DMV-Bench**, ahead of the best prior method at every setting on both backbones by 7.63 to 22.62 points; on MemLens it leads the strongest published memory-augmented agent by at least 8.20 points at all four context lengths, on a tenth to a seventieth of the full-context input. At MemLens 32K with dev-fitted selection it **surpasses GPT-5.4 and matches Gemini-3.1-Pro** while delivering about 2,400 tokens, a tenth of what they read.

Business project: LLM post-training for in-cabin AI (sole owner)
- **SFT**: LoRA fine-tuning on Qwen3-4B / 8B and Qwen3.6-35B-A3B (MoE) with ms-swift; implemented a set-prediction loss (minimum-permutation CE), hard-case weighted SFT and rejection-sampling fine-tuning (RFT).
- **Preference optimization and RL, with reward design**: DPO and triplet-level DPO (preference pairs split by Hungarian alignment), GRPO / DAPO; designed a hierarchical multi-component GRPO reward (format validity, schema, hierarchical triplet F1, direction, relation class and process rewards).
- **Evaluation**: built a triplet evaluator (Hungarian matching, placeholder equivalence, relation-synonym normalization, per-class and per-step reporting) with batched vLLM inference; audited and repaired the business memory evaluation set, replacing LLM-based pairing with a deterministic memory-level metric.

Research tracking: responsible for following new papers at top venues such as EuroSys and ICML and presenting them internally.

Fondazione Bruno Kessler - FBK Mar 2025 - Mar 2026
Research Intern Trentino-Alto Adige, Italy
- **Proposed Active Instance Verification and built its first benchmark, PInVerify.** Navigation benchmarks count arrival as success, yet an agent can stop in front of the wrong instance of the right category. Formalized this stage as a finite-horizon decision process and built 3,000 evaluation episodes over 18 object categories in photorealistic 3D scenes, with trap views and unreachable sectors in the navigation topology.

- **Designed the evaluation protocol and failure diagnostics.** Positive, same-category and cross-category pairs are balanced 1:1:1 to separate calibration bias from category recognition; beyond accuracy, the protocol reports steps to decision and navigation failure rate, and the training pool is disjoint from the test set. Judging differences by 95% confidence intervals located the main bottleneck in detection (+3.1 pp with ground-truth boxes) and found no reliable difference among three next-best-view strategies.
- **Built two baseline tracks, training-free and trained.** The training-free track is a modular pipeline (attribute decomposition, multi-view tracking and fusion, three next-best-view strategies), evaluated on Qwen3-VL-4B / 8B, SenseNova-SI-InternVL3-8B, CLIP and SigLIP2; the trained track is an end-to-end Qwen3-VL-4B + LoRA agent that outputs the verification decision and the next action in one forward pass.
- **Built a CoT dataset and post-trained the agent.** Generated about 22K CoT samples from benchmark ground truth and per-step visibility metadata, then compared DPO, GRPO and GSPO on top of SFT with a reward combining verification correctness, action quality and format. Accuracy rose from 70.6% untuned to 84.8% with SFT and **85.6%** with SFT+GSPO (88.9% with ground-truth boxes), and adding either GRPO or GSPO on top of SFT cut navigation failures from 18.2% to about 9%.

Chengdu Jiaoda Guangmang Technology Co., Ltd. Jul 2022 - Sep 2022
Algorithm Intern Chengdu, Sichuan, China

- Researched the latest machine learning and industrial anomaly detection algorithms.
- Applied GAN models to reconstruct images to identify anomalies and evaluate the performance.
- Used the object detection neural network model to detect abnormal parts in high-speed rail supports.
- Assisted colleagues in developing a set of character recognition patents for LCD meters.

Publications

"GeoSelect: Spatial-Program Execution for Training-Free Referring Remote Sensing Image Segmentation"
IEEE Transactions on Geoscience and Remote Sensing (TGRS), 2026, IEEE Xplore · Project Page, arXiv:2607.03869
Reframes referring expression resolution as the execution of an explicit spatial program, covering the positional superlative, ordinal, and compositional expressions that continuous fields cannot. Reaches 58.86 mIoU on RRSIS-D and 55.27 on RISBench, more than twice and about 1.7 times the best prior training-free results.
Yuhang Jiang, Guohui Deng*, Miaozhong Xu, Chao Ruan, Jinling Zhao, Linsheng Huang

"The Scissors Effect: When Resize-Based Input Diversity Helps or Hurts Transfer Attacks"
Transactions on Machine Learning Research (TMLR), 2026, Project Page, arXiv:2606.22516, OpenReview
Shows that resize-based input diversity, applied by default in transfer attacks, reverses sign on adversarially trained surrogates. Robustness evaluations that apply it blindly are therefore optimistic about the models they test.
Yuhang Jiang, Xiaojing Chen

"PInVerify: An Offline Embodied Benchmark for Active Instance Verification"
Poster, **FMEA Workshop @ CVPR 2026**, Project Page, arXiv:2605.30639, OpenReview
Master's thesis project.
Proposes Active Instance Verification, a new embodied task, and builds the first benchmark for it. Finds that active viewpoint selection gives no reliable gain over choosing at random, and that detection quality, not verification logic, is what binds the task.
Yuhang Jiang

"When Dynamic Scaling Fails: A Scale–State Feedback Governs Recurrent-State Quantization in Linear-Attention and SSM LLMs"
LCFM Workshop @ NeurIPS 2026 (non-archival), Project Page, arXiv:2610.15139, OpenReview, Code
A per-step quantizer sits inside the recurrence, so scale and state feed back: INT4 that is lossless at the start destroys recall a thousand steps later, and containing the scale beats locating outliers.
Yuhang Jiang
*Note: * indicates the corresponding author; † denotes equal contribution.*

Preprints and Other Publications

"GeoScope: Training-Free Program Execution beyond the Counting Horizon for Ultra-High-Resolution Remote Sensing Question Answering"
arXiv:2611.12699 · co-first author · Project page · under review
Measures a counting horizon of about five objects per view, then splits views by that limit rather than a fixed grid: untrained, 16.3 points on XLRS-Bench and 36.2 on UHR-Micro regional counting.
Guohui Deng*†, **Yuhang Jiang**†, Chengjun Wang, Zhongwen Hu, Zhaocong Wu, Linsheng Huang, Jinling Zhao, Chao Ruan, Miaozhong Xu, Chi Chen

"Retrieved but Not Delivered: Multimodal Memory Delivery for Long-Term Agents"
arXiv:2609.32590 · first author · Project page · under review
The bottleneck of multimodal memory is delivery, not retrieval; training-free DeliverMem leads the strongest published methods on MemLens and DMV-Bench.
Yuhang Jiang, Qingwei Liao, Kaize Yin, Xingling Liu, Luca Cuomo, Silvio Bacci

"Median Temporal Ensembling: Training-Free Robust Aggregation for Action-Chunked Visuomotor Policies"
arXiv:2609.27167 · sole author · Project page · under review
Replacing the weighted mean of overlapping action chunks with a per-coordinate median is one line and needs no retraining: never worse across 25 settings for ACT and Diffusion Policy, significantly better in 15.

Yuhang Jiang
"A Circuit, Not The Circuit: Non-Unique Causal Localisation of the Mamba-2 State Sink"
arXiv:2606.00930
Names the state sink, the state-space analogue of the attention sink. A near-disjoint set of heads matches the causal effect of the probe-selected specialists, and the effect dissociates by intervention surface, so a probe recovers a circuit rather than the circuit.
Yuhang Jiang, Bowen Zhang
"Refit the Probe: Single-Direction Ablation Is Not a Necessity Test"
arXiv:2606.00926
Ablating the direction a probe finds does not remove the target: a probe refitted on the ablated activations recovers its original accuracy, and keeps recovering when the probe's entire row space is deleted, because the quantity survives in the orthogonal complement. Neither a large task drop nor a near-zero one therefore establishes necessity. Replacing the ablation with iterative nullspace projection, scored against random subspaces of matched dimension, reverses the conclusion: representations that looked causally inert carry most of the task.
Yuhang Jiang
Software Copyright, Intelligent contract software for agricultural insurance compensation, 2022SR1568111.

Academic Service

Reviewer, ICLR 2027, ICRA 2027.
Reviewer, FMEA (Foundation Models Meet Embodied Agents) Workshop @ CVPR 2026.

Selected Projects

Research on Semantic Segmentation Method of High-Resolution Remote Sensing Images May 2023
Outstanding Undergraduate Graduation Project
- Developed an encoder-decoder model with residual-weighted attention for remote sensing segmentation, achieving F1-scores of 90.23% and 87.37% on ISPRS Potsdam and Vaihingen datasets.

Other projects. Multi-objective flexible job-shop scheduling with an enhanced NSGA-II (2025); HGT-GWO, a parallel Grey Wolf Optimizer in C/MPI/OpenMP on UNITN's HPC cluster (2025); AR-based robotic arm control on HoloLens 2 with YOLOv8 and FastSAM (2024).

Competitions

Kaggle March Machine Learning Mania 2025 - Silver Medal (64th / 1,727) Apr 2025
Solo

Kaggle BirdCLEF 2024 Competition - Bronze Medal (70th / 974) Jun 2024
Solo

Kaggle BirdCLEF+ 2025 Competition - Bronze Medal (199th / 2,025) Jun 2025
Solo

Mathematical Contest In Modeling (MCM), Finalist winner (top 1%) Apr 2021
Leader
- Designed a set of algorithms to assess the degree of hunger around the world and optimize the food supply chain.
- Responsible for modeling, programming and part of paper writing.
- Youngest winner in the history of Anhui University; first winner from the School of Internet.

iCAN Innovation Contest 2021 (Anhui) - Second prize Oct 2021
Team member
- Developed a set of hardware system capable of intelligently monitoring and warning the abnormal behavior of the elderly.
- Responsible for some STM32 development and project plan writing.

Skills

- **Programming:** Python, C/C++, Matlab, R.
- **ML / DL Frameworks:** PyTorch, scikit-learn; MPI / OpenMP (HPC).
- **LLM / VLM:** multimodal LLMs (MLLMs), post-training (SFT, DPO, GRPO / GSPO / DAPO, RFT), LoRA / PEFT, RAG, LLM agents.
- **Architectures:** Transformer, Mamba, CNN, RNN.
- **Applications:** Image Classification & Segmentation, Industrial Anomaly Detection, Time-Series Analysis.
- **Other:** Unity; Lean Manufacturing & Sustainable Management.
- **Languages:** English (fluent, TOEFL 102); Chinese (native).